supernova, and it produces a kind of characteristic rising and falling of brightness and a telltale spectrum of light that we can pretty reliably distinguish from other cosmic conflagrations. In principle, if you understand the physics of this kind of explosion really well, you know how bright it would be up close, and factoring in how bright it looks from all the way out here, you can deduce how far the light has traveled. (We call this the "standard candle" method because it's like you have a light bulb where you know the exact wattage. Once you have that information, you can always deduce the distance using the fact that the bulb will look dimmer when it's far away by a factor of the distance squared. Only we say "candle" instead of "light bulb" because it sounds more poetic that way.)

Once you have a measure of the distance, you need to know how fast the supernova is receding. For that, you can use the redshifting of the light from the galaxy the star exploded in, which tells you how quickly cosmic expansion is happening at that point. Use the distance and the speed of light to work out how long ago this whole thing went down, and you have a measurement of the expansion rate in the past.

In 1998, just a few years after that *Discover Magazine* article raising the alarm about the age of the cosmos, two independent research groups collecting observations of distant supernovae came to the same utterly unreasonable conclusion. That deceleration parameter—the one measuring how quickly the expansion rate was slowing down—was negative. The expansion wasn't slowing down at all. It was speeding up.

THE SHAPE OF THE COSMOS

If the cosmos were behaving itself, the basic physics involved in the expansion of the universe should be about as simple as throwing a ball up into the air, as we discussed in the previous chapter. Throw it too slowly, it goes up for a bit, slows down, stops, and falls down again: that's like a universe where there's enough matter (or a weak enough initial Big Bang expansion) that gravity wins and recollapses the universe. Throw the ball *incredibly inhumanly fast* and it might just escape the Earth's pull and drift out in space forever, always slowing: a universe perfectly balanced between expansion and gravity. Throwing it even faster than that means it'll escape and just coast forever, approaching a constant speed as the gravity of Earth becomes less and less of an influence: that's like a universe that keeps